CLASS 10 · CBSE · NCERT MATHEMATICS · DPP SHEET

Chapter 1: Real Numbers

Daily practice problems: 10 questions with an answer key

Try every question on your own first. The answer key is on the last page.

1. Express 5005 as a product of its prime factors.

2. Find the HCF and LCM of 252 and 90 by prime factorisation.

3. The HCF of two numbers is 23 and their LCM is 1449. If one number is 161, find the other.

4. Can 15^n end with the digit 0 for any natural number n? Give a reason.

5. Explain why $3 \times 5 \times 7 \times 11 + 7$ is a composite number.

6. Find the HCF and LCM of 15, 16 and 49.

7. Two tankers contain 850 litres and 680 litres of oil. Find the greatest capacity of a container that can measure the oil of both tankers an exact number of times.

8. Three traffic lights change after every 48, 72 and 108 seconds. If they change together at 7:00:00 a.m., when will they next change together?

9. Prove that $\sqrt{3}$ is irrational.

10. Prove that $5 - 2\sqrt{3}$ is irrational, given that $\sqrt{3}$ is irrational.

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Answer Key

1. $5005 = 5 \times 7 \times 11 \times 13$
2. $252 = 2^2 \times 3^2 \times 7$, $90 = 2 \times 3^2 \times 5$; HCF $= 18$, LCM $= 1260$
3. $\dfrac{23 \times 1449}{161} = 207$
4. No. $15^n = 3^n \times 5^n$ has no factor 2.
5. It equals $7\,(165 + 1) = 7 \times 166$, so it has factors other than 1 and itself.
6. HCF $= 1$, LCM $= 11760$
7. HCF$(850, 680) = 170$ litres
8. LCM $= 432$ s $= 7$ min 12 s, so at 7:07:12 a.m.
9. Assume $\sqrt{3} = \frac{a}{b}$ with a, b co-prime. Then $a^2 = 3b^2$, so 3 divides a; put $a = 3c$ to get $b^2 = 3c^2$, so 3 divides b. This contradicts co-primality.
10. If $5 - 2\sqrt{3} = \frac{a}{b}$, then $\sqrt{3} = \frac{5b-a}{2b}$, which is rational: a contradiction.

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